

NUMERICAL INTEGRATION AND DIFFERENTIATION

Outline

- Motivation
- Numerical Integration
- Numerical differentiation

Motivation

- Numerical Integration
 - Analytical integration cannot be performed
 - computation of mean
 - engineering applications:
 - volume of a reservoir, area of a watershed, mass from flux, marginal PDF from joint PDF
 - Solution to some Ordinary differential equations
 - Finite element method element matrix computation
- Numerical Differentiation
 - Finite-differences
 - Gradient computation
 - Groundwater flow fields
 - Jacobian computation in many numerical methods

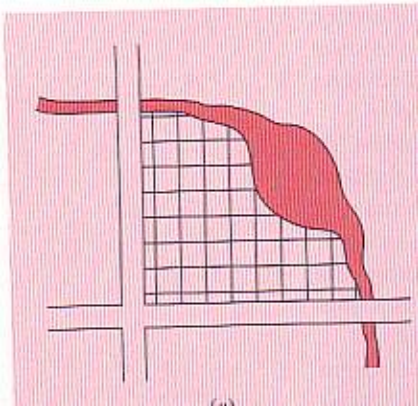
NUMERICAL INTEGRATION

Numerical Integration Outline

- Introduction
- Newton's cotes formulas
 - Trapezoidal Rule
 - Simpson's 1/3 rule
 - Simpson's 3/8 rule
- Gauss Quadrature
- Monte-Carlo Integration
- Extension to Multiple Integrals
- Integrals with variable limits
- Improper Integrals
- Engineering problems

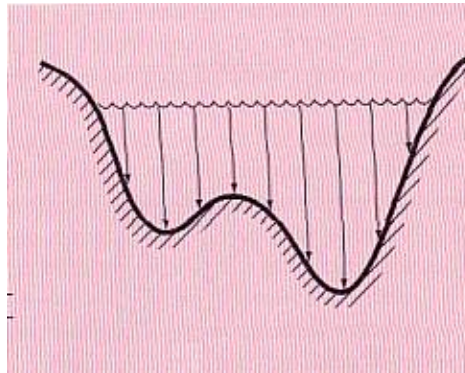
Introduction

Engineering Examples



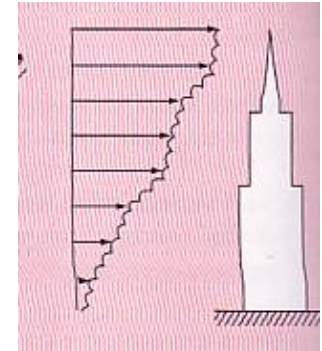
Area of a field

$$A = \iint_A dA = \sum_{i=1}^n \Delta A_i$$



Volume of a reservoir

$$V = \iiint_V dV = \sum_{i=1}^n \Delta V_i$$



Wind force on
a building

$$F = \int_{dh} f dh = \sum_{i=1}^n f_i \Delta h_i$$

Types of Integrals

- Line integrals
- Area integrals
- Volume integrals
- definite and indefinite integrals
- improper integrals

Types of Integrands

- Simple continuous function
- Complicated continuous function
- Tabulated function or data

Bungee Jumper Problem (Chapra p.5)

- ODE

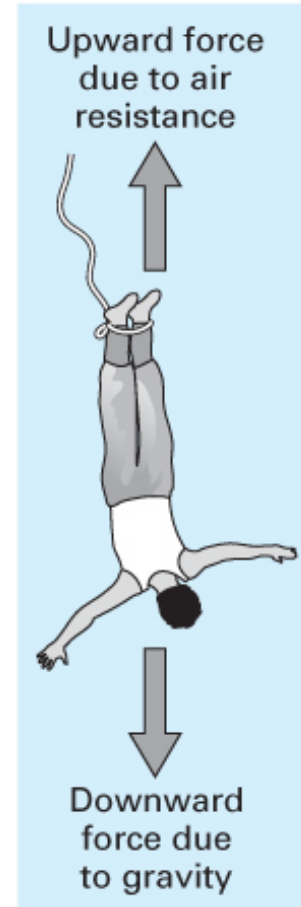
$$\frac{dv}{dt} = g - \frac{c}{m}v^2$$

- Solution

$$v(t) = \sqrt{\frac{gm}{c}} \tanh\left(\sqrt{\frac{gc}{m}}t\right)$$

- Distance traveled in time T from rest given by line integral

$$d = \int_0^T \sqrt{\frac{gm}{c}} \tanh\left(\sqrt{\frac{gc}{m}}t\right) dt$$



Line Integrals

- Summation of $f(x)dx$ over the range $x=a$ to b .

$$I = \int_a^b f(x)dx$$

- Mean of a continuous function in the interval a to b

$$mean = \frac{\int_a^b f(x)dx}{b-a}$$

Area Integrals

- Area of a water shed

$$A = \iint_A dA = \sum_{i=1}^n \Delta A_i$$

- Average of a two-dimensional function

$$average = \frac{\int_c^d \left(\int_a^b f(x, y) dx \right) dy}{(d - c)(b - a)}$$

- Heat or mass transfer (amount = flux x area)

$$\begin{aligned} \text{heat or mass transfer} &= \iint_A (\text{flux}) dA \\ &= \sum_{i=1}^n \text{flux}_i \Delta A_i \end{aligned}$$

Volume Integrals

- Mass computation (mass = concentration x volume)

$$\begin{aligned} mass &= \iiint_V c(V) dV \\ &= \iiint c(x, y, z) dx dy dz \\ &= \sum_{i=1}^n c_i \Delta V_i \end{aligned}$$

Other Definitions

- Definite Integrals – integrals with upper and lower limits

$$I = \int_a^b f(x)dx$$

- Indefinite integrals – integrals with no limits

$$I = \int f(x)dx$$

- Improper integrals – integrals with $-\infty$ or $+\infty$ as lower or upper limits

$$I = \int_{-\infty}^{\infty} f(x)dx$$

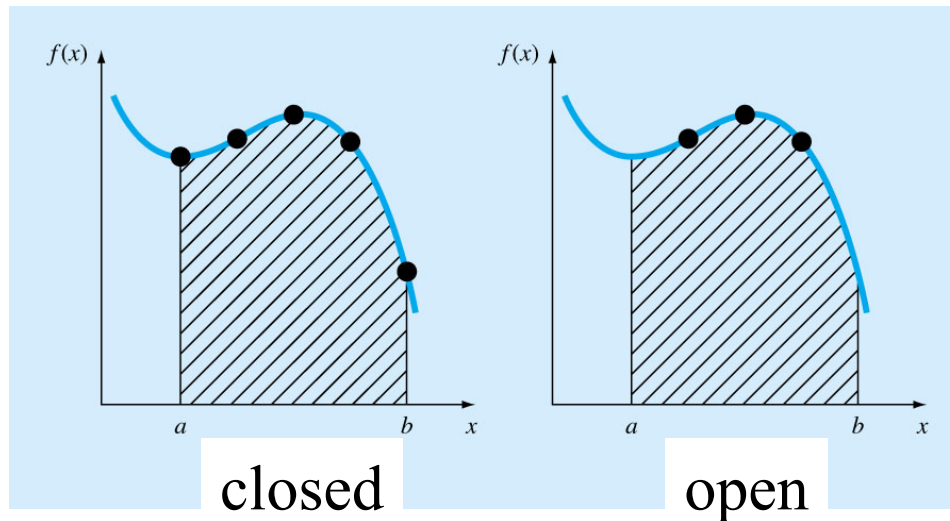
Newton-Cotes Formulas

Newton-Cotes Formulas

- The integral is approximated by a series of polynomials applied piecewise to the function or data over segments of constant length.
- Trapezoidal rule
- Simpson's $1/3$ rule
- Simpson's $3/8$ rule

Open and Closed Formulas

- Closed Newton-Cotes formulas are commonly used for evaluating definite integrals
- Open formulas are only used for improper integrals and solution of ODEs



Trapezoidal Rule

- The integral is approximated by

$$I = \int_a^b f(x) dx$$

$\cong$ width $\times$ average height

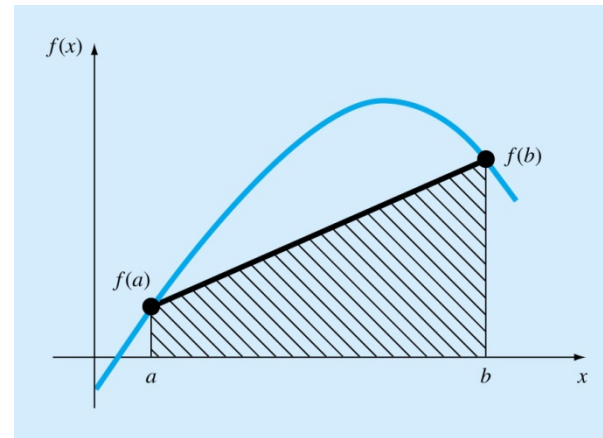
$$= (b-a) \frac{f(a) + f(b)}{2}$$

$$I = (b-a) \frac{f(a) + f(b)}{2}$$

- Error

$$\text{error} = -\frac{1}{12} f''(\xi)(b-a)^3$$

$$a < \xi < b$$



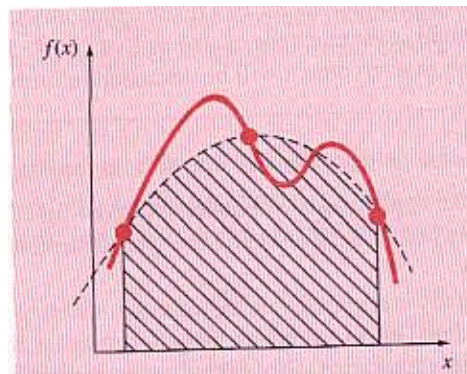
Simpson's 1/3 Rule

$$I = \int_a^b f(x) dx$$

$$\square \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$$\text{where } x_0 = a, \quad x_1 = \frac{a+b}{2}, \quad x_2 = b, \quad h = \frac{b-a}{2}$$

$$\text{error} = -\frac{(b-a)^5}{2880} f^{(4)}(\xi)$$



$$I = h \left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{3} \right]$$

$$h = \frac{b-a}{2}$$

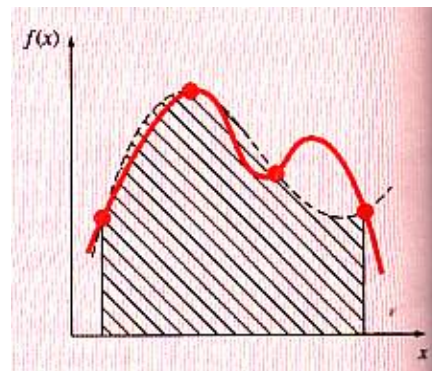
Simpson's 3/8 Rule

$$I = \int_a^b f(x) dx$$

$$\square \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

$$\text{where } x_0 = a, \quad h = \frac{b-a}{3}, \quad x_1 = a + h, \quad x_2 = a + 2h, \quad x_3 = b$$

$$\text{error} = -\frac{(b-a)^5}{6480} f^{(4)}(\xi)$$



$$I = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

$$h = \frac{b-a}{3}$$

Derivation of Newton-Cotes Formulas

- The formulas can be derived from Newton's interpolating polynomials to the integrand (i.e., $f(x)$).

$$f(x) = f(x_0 + \alpha h) = f_0 + \alpha \Delta f_0 + \frac{\alpha(\alpha-1)\Delta^2 f_0}{2!} + \cdots + E_n(\alpha)$$

$$E_n(\alpha) = \frac{\alpha(\alpha-1)\cdots[\alpha-(n-1)]f^{(n)}(\xi)h^n}{n!}$$

Note: $h = b - a$; α varies from 0 to 1

Newton's Forward differences of $f(x)$ at $x=x_0$

k	$\Delta^k f_0$
1	$f_1 - f_0$
2	$f_2 - 2f_1 + f_0$
3	$f_3 - 3f_2 + 3f_1 - f_0$
4	$f_4 - 4f_3 + 6f_2 - 4f_1 + f_0$
5	$f_5 - 5f_4 + 10f_3 - 10f_2 + 5f_1 - f_0$
6	$f_6 - 6f_5 + 15f_4 - 20f_3 + 15f_2 - 6f_1 + f_0$

Derivation of Newton Cotes - Trapezoidal

Letting $x = x_0 + \alpha h$ and $dx = h d\alpha$,

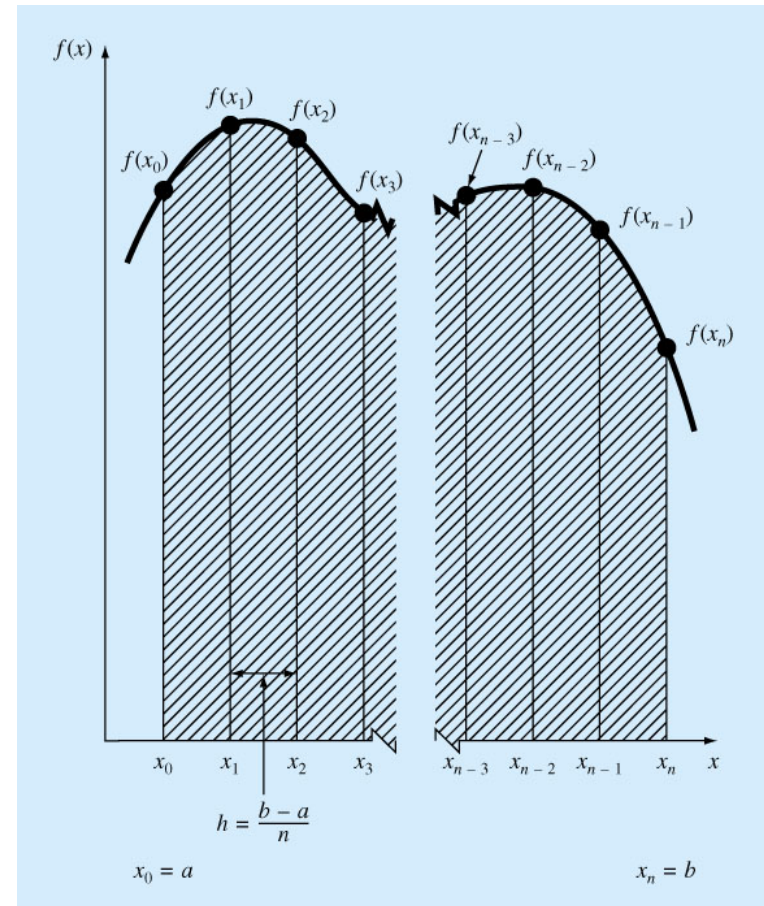
$$\begin{aligned} I(x_0, x_1) &= \int_{x_0}^{x_1} f(x) dx = \int_0^1 [f_0 + \alpha \Delta f_0 + E_n(\alpha)] h d\alpha \\ &= \int_0^1 [f_0 + \alpha(f_1 - f_0)] h d\alpha + \int_0^1 \frac{\alpha(\alpha - 1)f^{(2)}(\xi)h^2}{2!} h d\alpha \\ &= \frac{h(f_0 + f_1)}{2} - \frac{f^{(2)}(\xi)h^3}{12} \end{aligned}$$

Trapezoidal Rule

Note: $h = x_1 - x_0$; α varies from 0 to 1

Multiple Applications of Newton-Cotes formulas

- The integration interval can be divided into several small equal segments and the Newton-Cotes formulas can be applied to these segments.
- The number of segments is the total number of function evaluations less one.
 - For example, two applications of Simpson's 1/3 rule results in four segments.

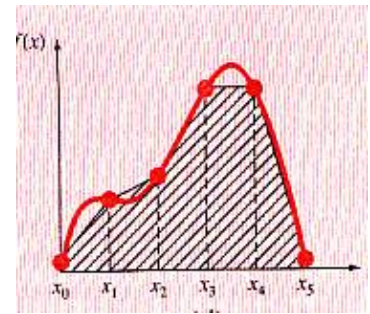


Composite Trapezoidal Rule

- The integral is approximated using multiple applications of the trapezoidal rule

$$\begin{aligned} I &= \int_a^b f(x)dx = \int_{x_0}^{x_1} f(x)dx + \int_{x_1}^{x_2} f(x)dx + \cdots + \int_{x_{n-1}}^{x_n} f(x)dx \\ &= h \frac{f(x_0) + f(x_1)}{2} + h \frac{f(x_1) + f(x_2)}{2} + \cdots + h \frac{f(x_{n-1}) + f(x_n)}{2} \end{aligned}$$

$$\text{where } x_0 = a, \quad x_n = b, \quad h = \frac{b-a}{n}$$



$$I = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

$$h = \frac{b-a}{n}$$

Composite Simpson's 1/3 Rule

$$I = \frac{h}{3} \left[f(x_0) + 4 \sum_{i=1,3,5}^{n-1} f(x_i) + 2 \sum_{i=2,4,6}^{n-2} f(x_i) + f(x_n) \right]$$

$$h = \frac{b - a}{n}$$

Total number of segments n is even

Composite Simpson's 3/8 Rule

$$I = \frac{3h}{8} \left[f(x_0) + 3 \sum_{i=1,4,7}^{n-3} f(x_i) + 3 \sum_{j=2,5,8}^{n-2} f(x_j) + 2 \sum_{k=3,6,9}^{n-1} f(x_k) + f(x_n) \right]$$

$$h = \frac{b-a}{n}$$

Total number of segments n is a multiple of 3

Implementation

- Trapezoidal rule is the most commonly used of the 3 Newton-Cotes formulas.
- Simpson's $1/3$ and $3/8$ rules have almost the same amount of accuracy but $1/3$ rule uses fewer number of points.
- If the number of segments (n) is odd, generally $1/3$ rule is applied for the first $n-3$ even number of segments and $3/8$ is applied for the last three segments.
- In general, multiple segment Newton-Cotes formulas are implemented with the number of segments and the integration limits passed as arguments.
 - Multiple segment Newton-Cotes can also be implemented by repeatedly calling the single segment versions – but this is less preferred as it involves additional overhead

MATLAB Example for Line Integration

[line integral example.m](#)

- Function:

$$I = \int_0^3 x e^x dx$$

- Trapezoidal rule
- Simpson's rule
- Analytical solution (`int` symbolic function)
- MATLAB function `integral`
- Inclass Example:
 - [Bungee Jumper Line Integral](#)

$$d = \int_0^T \sqrt{\frac{gm}{c}} \tanh\left(\sqrt{\frac{gc}{m}} t\right) dt$$

Summary of Newton-Cotes Closed Formulas

Segments (n)	Points	Name	Formula	Truncation Error
1	2	Trapezoidal rule	$(b-a) \frac{f(x_0) + f(x_1)}{2}$	$-(1/12)h^3 f''(\xi)$
2	3	Simpson's 1/3 rule	$(b-a) \frac{f(x_0) + 4f(x_1) + f(x_2)}{6}$	$-(1/90)h^5 f^{(4)}(\xi)$
3	4	Simpson's 3/8 rule	$(b-a) \frac{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)}{8}$	$-(3/80)h^5 f^{(4)}(\xi)$
4	5	Boole's rule	$(b-a) \frac{7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)}{90}$	$-(8/945)h^7 f^{(6)}(\xi)$
5	6		$(b-a) \frac{19f(x_0) + 75f(x_1) + 50f(x_2) + 50f(x_3) + 75f(x_4) + 19f(x_5)}{288}$	$-(275/12,096)h^7 f^{(6)}(\xi)$

Integration of Tabular Data

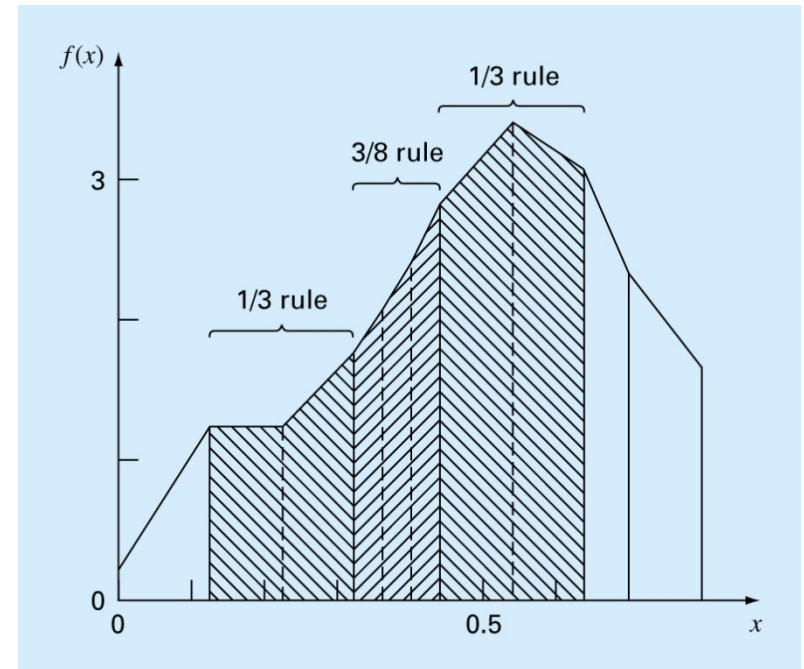
- Most experimental data is in tabular form.
- Tabular data can have equal or unequal segments.
- If they have unequal segments then the integral can be evaluated using trapezoidal rule as follows:

$$I = h_1 \frac{f(x_0) + f(x_1)}{2} + h_2 \frac{f(x_1) + f(x_2)}{2} + \dots + h_n \frac{f(x_{n-1}) + f(x_n)}{2}$$

where $h_i = x_i - x_{i-1}$ = width of segment i

Integration of Tabular Data (contd.)

- If they have equal size segments then Simpson's $1/3$ or $3/8$ rules can be applied.
- If they have a combination of equal and unequal segments then trapezoidal rule can be applied to unequal segments and Simpson's rules can be applied to equal segments.



Integration of Tabular Data (contd.)

- Tabular data can also be integrated by using a large number of equal segments and using *interpolation* to evaluate each point.
- But this may result in inaccurate results if the data is highly irregular.
- In MATLAB interpolations can be performed to various orders using one of the following functions: `interp1` (for 1D data) `interp2` (for 2D data), `interp3` (for 3D data), or `interp` (for n-d data).
- Tabular data can be incorporated into a function.

Tabular Data Example 1

C & C 21.23 (p. 631)

- Calculate distance 'd' from velocity data with time
 - Method1: Trapezoidal rule for tabular data
 - Method2: linearly interpolate and integrate using 'trap' function

$$d = \int_{t_1}^{t_2} v(t) dt = \sum_i v_i \Delta t_i$$

t (min)	0	1	2	3.25	4.5	6	7	8	8.5	9	10
v (m/s)	0	5	6	5.5	7	8.5	8	6	7	7	5

Tabular data example 2

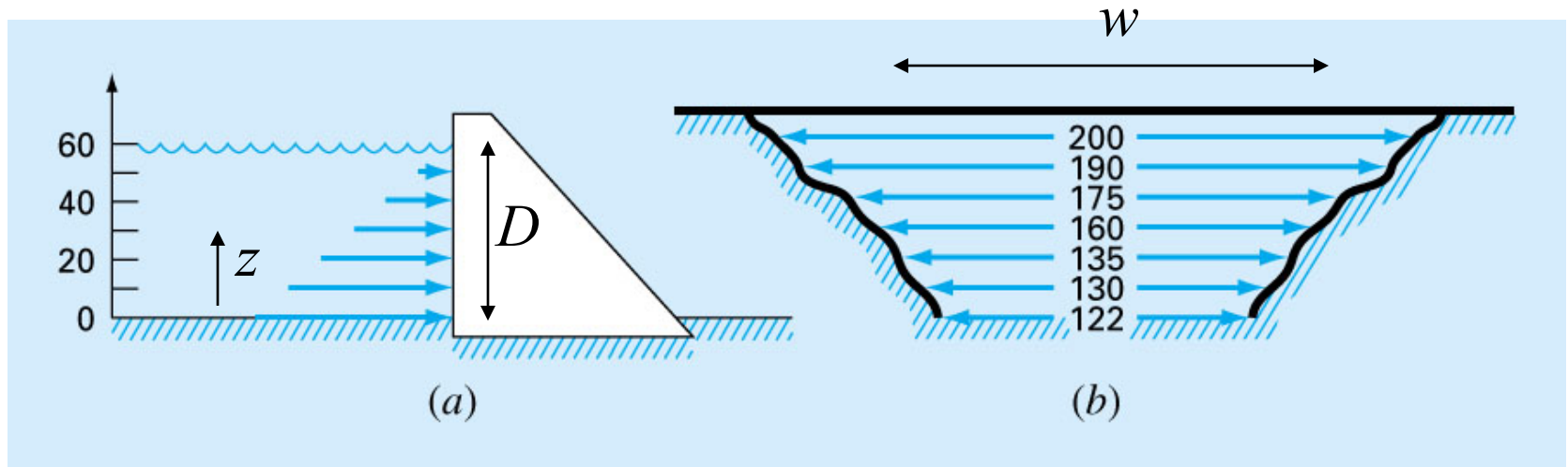
(C & C 24.5 p. 684)

- Mass leaving a reactor

$$M = \int_{t_1}^{t_2} Qcdt = \sum_i Q_i c_i \Delta t_i$$

t (min)	0	10	20	30	35	40	45	50
Q (m ³ /min)	4.0	4.8	5.2	5.0	4.6	4.3	4.3	5.0
C (mg/m ³)	10	35	55	52	40	37	32	34

Tabular Data Example 3 (C&C 24.23, p 688)

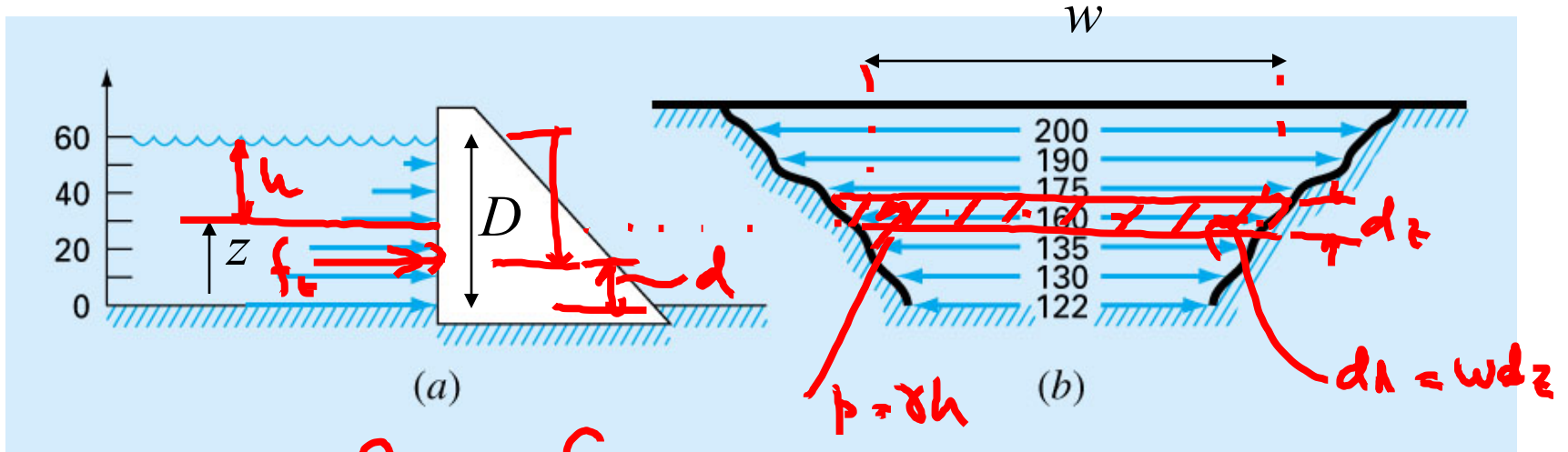


$$f_t = \int_0^D \rho g (D - z) w(z) dz$$

$$d = \frac{\int_0^D \rho g z (D - z) w(z) dz}{\int_0^D \rho g (D - z) w(z) dz}$$

Dam example (contd.)

Tabular Data Example 3 (C&C 24.23, p 688)



$$F = \int p \, dA = \int_{\text{bottom}} \rho g h w \, dz$$

$$f_t = \int_0^D \rho g(D-z)w(z)dz$$

$$d = \frac{\int_0^D \rho g z (D - z) w(z) dz}{\int_0^D \rho g (D - z) w(z) dz}$$

Tabular example 4 (C & C 20.20 p. 631)

- Use numerical integration to calculate weighted average concentration leaving the reactor over a 24-hr period
- Outflow concentration data

T (hr)	0	1	5.5	10	12	14	16	18	20	24
C (mg/L)	1	1.5	2.3	2.1	4	5	5.5	5	3	1.2

- Flow rate equation (m^3/s): $Q(t) = 20 + 10 \sin\left(\frac{2\pi}{24}(t - 10)\right)$
- Average concentration given by:

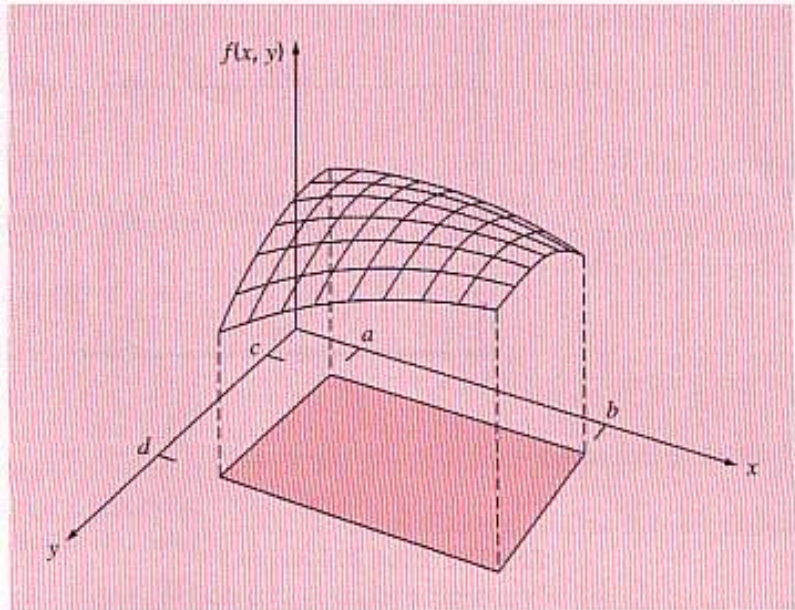
$$\bar{c} = \frac{\int_0^t Q(t)c(t)dt}{\int_0^t Q(t)dt}$$

Extension to Multiple Integrals

- Example – average of a 2D function

$$\text{average} = \frac{\int_a^b \left(\int_c^d f(x, y) dy \right) dx}{(b-a)(d-c)}$$

- Integrate in each dimension successively with values of the other dimensions held constant.
- Select a convenient order of integration (answer is the same regardless of the order).



Double Integral

- General function

$$I = \int_{ax}^{bx} \int_{ay}^{by} f(x, y) dy dx$$

- Treat the inner integral as a function to an outer line integral

MATLAB Example for Double Integral

double integral.m

- Function:

$$I = \int_{-1}^1 \int_0^3 x e^{xy} dx dy$$

- Trapezoidal rule
- Simpson's rule
- Analytical solution
- MATLAB function `dblquad` or `integral2`

Extension to higher number of dimensions

- The recursive calling functionality in MATLAB can be used to implement m-dimensional versions of Trapezoidal and Simpson's formulas.
- The limits and the number of segments are given as row vectors.
- Example codes:
 - Driver function: [mdim_example.m](#)
 - Trapezoidal function: [mtrap.m](#)
 - Simpson function: [msimp.m](#)

Some Notes on MATLAB implementation of higher dimensional integrations

$$I = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \cdots \int_{a_m}^{b_m} f(x_1, x_2, \cdots, x_m) dx_m dx_{m-1} \cdots dx_1$$

- MATLAB recursive functionality should be used
- Vectors (size = number of dimensions) should be used to represent integration limits (a_1, a_2 , etc), independent variables (x_1, x_2 , etc.), and number of segments in each direction (n 's).
- A convenient convention should be chosen
 - i.e, x_1 can be either the inner or outer variable (in the integral given above it is outer – same as matlab's integral function)

Tank Example 1

- Calculate the total contaminant mass in a rectangular tank given a non-uniform concentration distribution:

$$c(x, y, z) = \frac{C_0 (x^2 + y^2)}{\sqrt{z^2 + 1}}$$

- x, y are measured from the center and z is measured from the bottom
- Tank dimensions are 2 m x 2 m x 3 m and $C_0 = 100$ mg/L

Tank example 1 (contd.)

$$M = \int_{x=-1}^1 \left(\int_{y=-1}^1 \left(\int_{z=0}^3 \frac{C_0 (x^2 + y^2)}{\sqrt{z^2 + 1}} dz \right) dy \right) dx$$

Mass of Earth (c 20.25, p. 520)

- Calculate mass of earth using density data

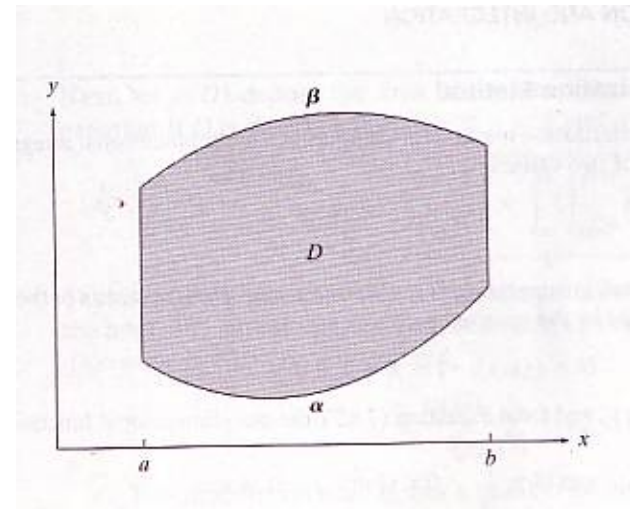
$$M = \int_0^{2\pi} \int_0^\pi \int_0^{6380} \rho r^2 \sin \alpha dr d\alpha d\beta = \int_0^{6380} 4\pi \rho r^2 dr$$

r (km)	0	1100	1500	2450	3400	3630	4500	5380	6060	6280	6380
ρ (Mg/m ³)	13	12.4	12	11.2	9.7	5.7	5.2	4.7	3.6	3.4	3

Extension to Variable Limits

- Multiple integrals can sometimes have inner limits expressed as functions of the outer variable:

$$I = \int_{x=a}^{x=b} \left(\int_{y=\alpha(x)}^{y=\beta(x)} f(x, y) dy \right) dx$$



MATLAB implementation of Integration with functions as limits

- Integrations with functions as limits (variable limit integrals) can be implemented in Matlab by calling the appropriate function each time a limit is to be evaluated.
- Examples:

$$I = \int_{x=0}^5 \left(\int_{y=0}^x xy dy \right) dx$$

$$I = \int_{x=0}^8 \left(\int_{y=0}^{x/2} \left(\int_{z=0}^{x-2y} e^{-(x^2+y^2+z^2)/4} dz \right) dy \right) dx$$

MATLAB Implementation (contd.)

- MATLAB functions `integral2` or `integral3` *now accepts* functions as integration limits.
- A general variable limit m-dimensional integration function can be implemented but this requires some work.
- Limits need to be passed in as *character arrays or cell arrays*.
- If using character arrays
 - Use `str2num` function to convert constant limits to values
 - Use `dblank` function to delete blanks at the end of function names
- If using cell arrays (preferred as it works with handles)
 - Use `isnumeric` function to check whether it is a number or a function
- Examples:
 - Cell array version: [variable limit example.m](#) and [gtrap.m](#) and [gsimp.m](#)
 - Character array version: [general integral example.m](#)

Tank Example 2

- Extend previous tank example with time dependent change in fluid volume
- Assume there is a volumetric influx of liquid into the tank at the following sinusoidal rate:

$$Q = Q_0 \left(1 + \sin(\omega t) \right)$$

- Assume the tank is empty in the beginning and the concentration distribution is the same as before
- Use $Q_0 = 100 \text{ L/s}$, $\omega = \pi/6$, t is in seconds
- Calculate total mass at $t = 1 \text{ hour}$
 - tank_mass_fixed_time_dependent_height_no_decay.m

Tank example 2 (contd.)

$$M = \int_{x=-1}^1 \left(\int_{y=-1}^1 \left(\int_{z=0}^{H(t)} \frac{C_0 (x^2 + y^2)}{\sqrt{z^2 + 1}} dz \right) dy \right) dx$$

$$H(t) = \frac{\int_0^T Q_0 (1 + \sin(\omega t)) dt}{A}$$

Tank Example 3

- Extend previous example for time dependent concentration in tank with the following distribution

$$c(x, y, z, t) = \frac{C_0 e^{-kt} (x^2 + y^2)}{\sqrt{z^2 + 1}}$$

- Assume the tank is empty in the beginning
- Use $k = 0.001$, all other parameters remain same
- Plot total mass (M) vs time (t) for $t = 0$ to 3600 s
- Calculate the time averaged mass in the tank over this period
 - tank_mass_average_time.m

Tank Example 3

$$M(T) = \int_{x=-1}^1 \left(\int_{y=-1}^1 \left(\int_{t=0}^T \left(\int_{z=0}^{H(t)} \frac{C_0 e^{-kt} (x^2 + y^2)}{\sqrt{z^2 + 1}} dz \right) dt \right) dy \right) dx$$

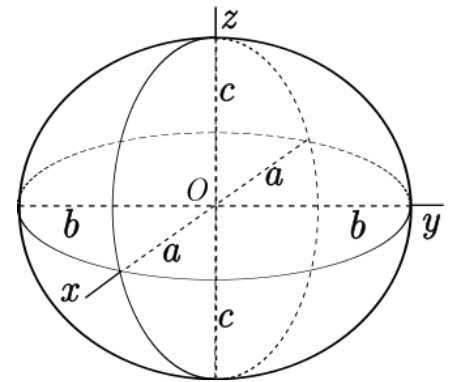
$$H(t) = \frac{\int_0^t Q_0 (1 + \sin(\omega t)) dt}{A}$$

$$\bar{M} = M(T) / T$$

Volume of an ellipsoid

[volume_ellipsoid.m](#)

$$V = \int_{x=-a}^{x=a} \int_{y=-b\sqrt{1-\frac{x^2}{a^2}}}^{y=b\sqrt{1-\frac{x^2}{a^2}}} \int_{z=-c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}}^{z=c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}} dz dy dx = \frac{4\pi abc}{3}$$

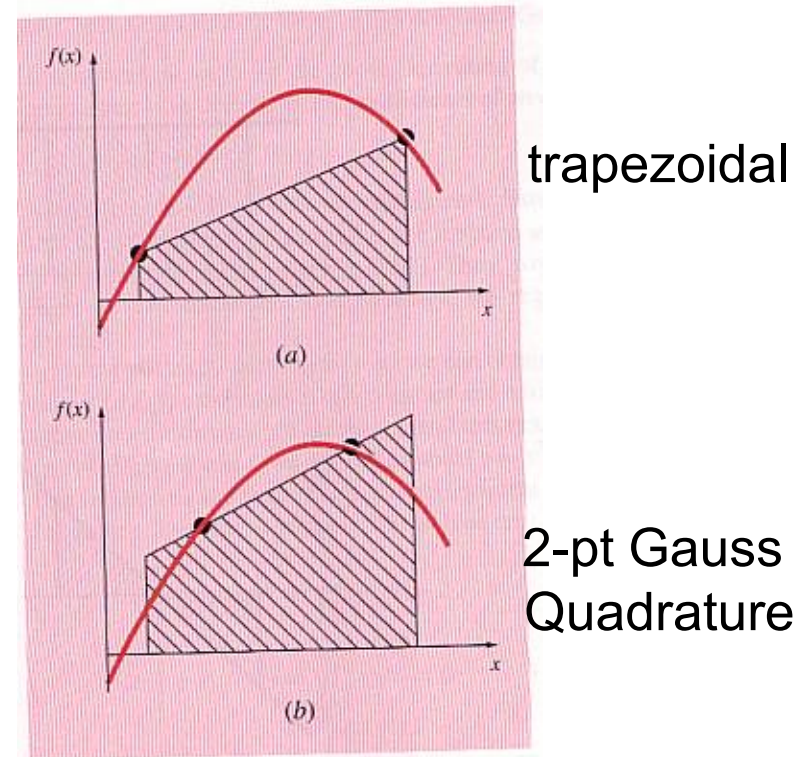


$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Guass Quadrature

Basics of GQ

- Gauss quadrature is similar to Newton-Cotes in that it evaluates the function to be integrated at several points and then sums the results.
- Gauss quadrature chooses points of evaluation judiciously instead of evenly spaced points in Newton-Cotes formulas.



Basics of GQ (contd.)

- Start from trapezoidal rule:

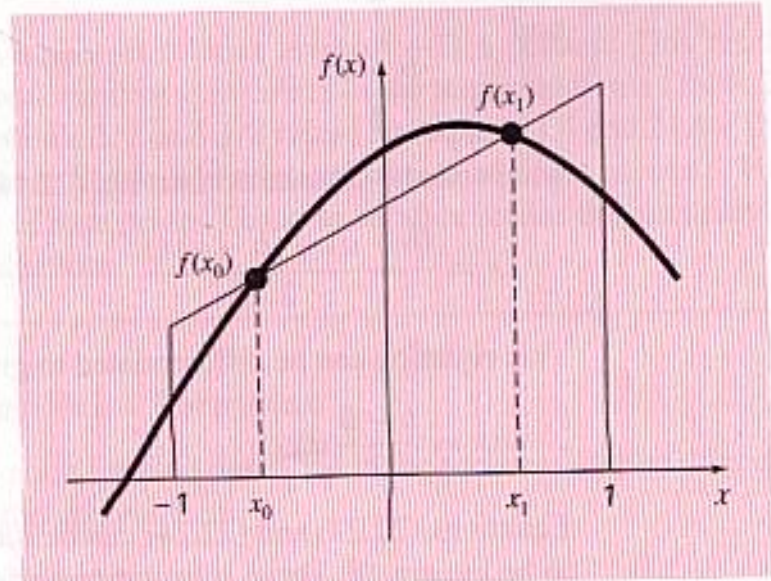
$$I = \int_a^b f(x)dx \approx (b-a) \frac{f(a) + f(b)}{2}$$

- What if we select two points between the limits s.t:

$$I = \int_a^b f(x)dx \approx c_0 f(x_0) + c_1 f(x_1)$$

- Or n points such that

$$I = \int_a^b f(x)dx \approx \sum_{i=0}^n c_i f(x_i)$$



Basics of GQ (contd.)

- The idea is to find weights c_i 's and points x_i 's such that the integral is exact for a polynomial of $2n-1^{\text{th}}$ degree.
- Typically the points and weights are expressed for an integral with limits $[-1,1]$.
- A simple linear transformation can then be used to convert to integrals with limits $[a, b]$.

2-pt GQ

$$\int_{-1}^1 f(x)dx = c_0 f(x_0) + c_1 f(x_1)$$

- 4 unknowns (c_0, c_1, x_0, x_1)
- We can find these 4 unknowns by requiring the integral to be exact for polynomials of degree 3 or less.
- Let us use the simplest cases. i.e., $f(x)=1, x, x^2, x^3$.

4 equations for 2-pt GQ

$$\left. \begin{aligned} \int_{-1}^1 1 dx &= 2 = c_0 + c_1 \\ \int_{-1}^1 x dx &= 0 = c_0 x_0 + c_1 x_1 \\ \int_{-1}^1 x^2 dx &= \frac{2}{3} = c_0 x_0^2 + c_1 x_1^2 \\ \int_{-1}^1 x^3 dx &= 0 = c_0 x_0^3 + c_1 x_1^3 \end{aligned} \right\}$$

$$c_0 = 1$$

$$c_1 = 1$$

$$x_0 = -\frac{1}{\sqrt{3}}$$

$$x_1 = \frac{1}{\sqrt{3}}$$

Note: we could use symmetry to determine that $c_0=c_1$ and $x_0=-x_1$

General GQ (C&C formulation)

$$\int_{-1}^1 f(x)dx \approx c_0 f(0) + \sum_{i=1}^{(n-1)/2} c_i (f(x_i) + f(-x_i)), \quad n \text{ is odd}$$
$$= \sum_{i=1}^{n/2} c_i (f(x_i) + f(-x_i)), \quad n \text{ is even}$$

General GQ (ignoring symmetry)

$$\int_{-1}^1 f(x) dx \approx \sum_{k=0}^N c_k f(x_k)$$

$$\int_{-1}^1 \int_{-1}^1 f(x, y) dx dy \approx \sum_{i=0}^N c_i \sum_{j=0}^N c_j f(x_i, y_j)$$

GQ for general limits

- Mapping limits $[-1, 1]$ to $[a, b]$ can be accomplished by the following linear transformation

$$\int_a^b f(x)dx = \alpha \int_{-1}^1 f(\alpha\xi + \beta)d\xi, \quad \text{where } x = \alpha\xi + \beta$$

$$\begin{array}{l} \alpha(-1) + \beta = a \\ \alpha(1) + \beta = b \end{array} \quad \Rightarrow \quad \begin{array}{l} \alpha = \frac{b-a}{2} \\ \beta = \frac{b+a}{2} \end{array}$$

GQ for general limits

$$\int_a^b f(x) dx \approx \alpha \sum_{k=0}^N c_k f(\alpha \xi_k + \beta)$$

$$\alpha = \frac{b-a}{2}, \quad \beta = \frac{b+a}{2}$$

How do we choose the number of Gauss points?

- If we want the integral to be exact for a polynomial of degree n or less then:
 - The number of Gauss points $N = (n+1)/2$

Gauss Point Table

Points	Weighting Factors	Function Arguments	Truncation Error
2	$c_0 = 1.0000000$ $c_1 = 1.0000000$	$x_0 = -0.577350269$ $x_1 = 0.577350269$	$\cong f^{(4)}(\xi)$
3	$c_0 = 0.5555556$ $c_1 = 0.8888889$ $c_2 = 0.5555556$	$x_0 = -0.774596669$ $x_1 = 0.0$ $x_2 = 0.774596669$	$\cong f^{(6)}(\xi)$
4	$c_0 = 0.3478548$ $c_1 = 0.6521452$ $c_2 = 0.6521452$ $c_3 = 0.3478548$	$x_0 = -0.861136312$ $x_1 = -0.339981044$ $x_2 = 0.339981044$ $x_3 = 0.861136312$	$\cong f^{(8)}(\xi)$
5	$c_0 = 0.2369269$ $c_1 = 0.4786287$ $c_2 = 0.5688889$ $c_3 = 0.4786287$ $c_4 = 0.2369269$	$x_0 = -0.906179846$ $x_1 = -0.538469310$ $x_2 = 0.0$ $x_3 = 0.538469310$ $x_4 = 0.906179846$	$\cong f^{(10)}(\xi)$
6	$c_0 = 0.1713245$ $c_1 = 0.3607616$ $c_2 = 0.4679139$ $c_3 = 0.4679139$ $c_4 = 0.3607616$ $c_5 = 0.1713245$	$x_0 = -0.932469514$ $x_1 = -0.661209386$ $x_2 = -0.238619186$ $x_3 = 0.238619186$ $x_4 = 0.661209386$ $x_5 = 0.932469514$	$\cong f^{(12)}(\xi)$

Implementation of Gauss Quadrature

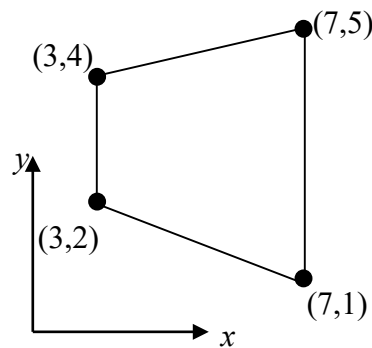
- Program for Gauss Quadrature can be implemented easily in MATLAB by storing the Gauss points and weights in a table.
- A general version can be implemented to handle multi-dimensional versions.
- Examples:
 - [simple_gauss.m](#)
 - [general_gauss_example.m](#) and [ggauss.m](#)

Use of Gauss Quadrature in FEM

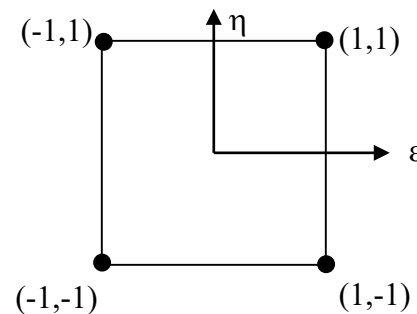
- GQ is commonly used in FEM local element matrix computation.

- Following is an expression for a quadrilateral element

$$[KM_e]_{4 \times 4} = \int_{-1}^1 \int_{-1}^1 [GN]_{4 \times 2} [J^{-1}]_{2 \times 2}^T [K_e]_{2 \times 2} [J^{-1}]_{2 \times 2} [GN]^T_{2 \times 4} |J| d\epsilon d\eta$$



Global Coordinates



Local Coordinates

GQ in FEM (contd.)

$$[GN]_{4 \times 2} = \begin{bmatrix} \frac{\partial N_1}{\partial \varepsilon} & \frac{\partial N_1}{\partial \eta} \\ \frac{\partial N_2}{\partial \varepsilon} & \frac{\partial N_2}{\partial \eta} \\ \frac{\partial N_3}{\partial \varepsilon} & \frac{\partial N_3}{\partial \eta} \\ \frac{\partial N_4}{\partial \varepsilon} & \frac{\partial N_4}{\partial \eta} \end{bmatrix}$$

$$N_i = \frac{1}{4}(1 + \varepsilon_i \varepsilon)(1 + \eta_i \eta)$$

$$\frac{\partial N_i}{\partial \varepsilon} = \frac{1}{4} \varepsilon_i (1 + \eta_i \eta)$$

$$\frac{\partial N_i}{\partial \eta} = \frac{1}{4} \eta_i (1 + \varepsilon_i \varepsilon)$$

$$\varepsilon_i = -1, 1, 1, -1 \quad \text{for } i = 1, 2, 3, 4$$

$$\eta_i = -1, -1, 1, 1 \quad \text{for } i = 1, 2, 3, 4$$

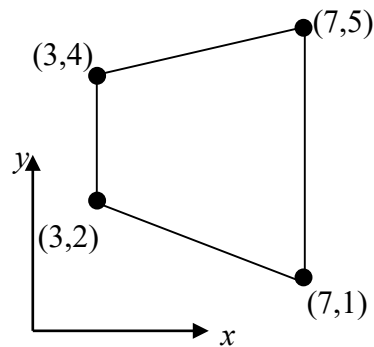
GQ in FEM (contd.)

$$[J]_{2 \times 2} = [GN]_{2 \times 4}^T \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix}_{4 \times 2}$$

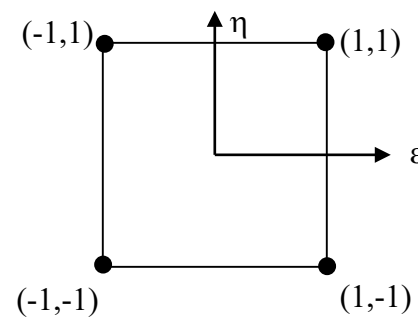
$$x_i = 3, 7, 7, 3 \quad \text{for } i = 1, 2, 3, 4$$

$$y_i = 2, 1, 5, 4 \quad \text{for } i = 1, 2, 3, 4$$

$$[K_e]_{2 \times 2} = \begin{bmatrix} K_x & 0 \\ 0 & K_y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$



Global Coordinates



Local Coordinates

Use of Gauss Quadrature in FEM

$$[KM_e]_{4 \times 4} = \int_{-1}^1 \int_{-1}^1 [GN]_{4 \times 2} [J^{-1}]_{2 \times 2}^T [K_e]_{2 \times 2} [J^{-1}]_{2 \times 2} [GN]^T_{2 \times 4} |J| d\varepsilon d\eta$$

- The above double integral can be approximated by

$$\int_{-1}^1 \int_{-1}^1 f(\varepsilon, \eta) d\varepsilon d\eta \approx \sum_{i=1}^n w_i \sum_{j=1}^n w_j f(\varepsilon_i, \eta_j)$$

- For $n = 2$ this reduces to

$$\int_{-1}^1 \int_{-1}^1 f(\varepsilon, \eta) d\varepsilon d\eta = f\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right) + f\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) + f\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$$

Monte-Carlo Integration

Basics

- Basic idea is to find average value of $f(x)$ over the interval $[a,b]$ and then multiply by $(b-a)$.
- The average value is found by calculating $f(x)$ at randomly generated points within interval $[a,b]$ summing and then dividing by the number of data points.

$$\int_a^b f(x)dx \approx \frac{(b-a)}{n} \sum_{i=1}^n f(x_i)$$

- The same concept can be extended to multiple integrals.
- Order of accuracy is only

$$1/\sqrt{n}$$

Monte-Carlo Integration for double and triple integrals

- Double integrals

$$\int_c^d \left(\int_a^b f(x, y) dx \right) dy \approx \frac{(b-a)(d-c)}{n} \sum_{i=1}^n f(x_i, y_i)$$

- Triple integrals

$$\int_e^f \left(\int_c^d \left(\int_a^b f(x, y, z) dx \right) dy \right) dz \approx \frac{(b-a)(d-c)(f-e)}{n} \sum_{i=1}^n f(x_i, y_i, z_i)$$

Extension of MC integration to complicated regions

- Monte-Carlo integration can also be performed for integrals with complicated domains.
- In these cases, the points (x_i, y_i, z_i) are randomly generated in a hyperrectangle containing the domain and then checked if they lie inside or outside the domain.
- If they are inside then they are evaluated and if they are outside then the value is set to zero.
- Examples:
 - [simple monte.m](#)
 - [general monte.m](#)

Improper Integrals

- Improper integrals are not uncommon
 - One or both limits are infinite (i.e., $-\infty$ or $+\infty$ as lower or upper limits).

$$I = \int_0^{\infty} \frac{4}{\pi(4+x^2)} dx$$

- Integrand “blows up” at the end points (singularity) or for some values within the limits.

$$I = \int_0^1 \frac{1}{\sqrt{x}} dx$$

- Integrals with discontinuous functions.
- Improper integrals can be evaluated by splitting integrals and then performing transformation of variables and then applying open or closed integration formulas.

Variable Transformation

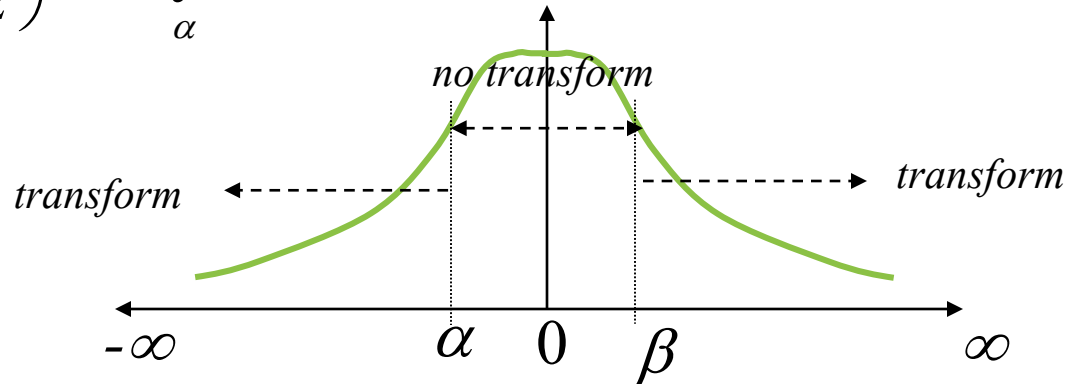
- For integrals with infinite limits, a variable transformation $x = 1/z$ is usually employed. In this case, $dx = -dz/z^2$ and the integral can be recast as:

$$I(a, b) = \int_a^b f(x) dx = \int_{1/b}^{1/a} \frac{1}{z^2} f\left(\frac{1}{z}\right) dz, \quad ab > 0$$

- If the integral has singularities at one or more end points then split the integral into 2 or 3 parts before applying variable transformation to satisfy ($ab > 0$)

Splitting and Variable Transformation

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^{\alpha} f(x) dx + \int_{\alpha}^{\beta} f(x) dx + \int_{\beta}^{\infty} f(x) dx \\
 &= - \int_0^{1/\alpha} \frac{1}{z^2} f\left(\frac{1}{z}\right) dz + \int_{\alpha}^{\beta} f(x) dx - \int_{1/\beta}^0 \frac{1}{z^2} f\left(\frac{1}{z}\right) dz \\
 &= \int_{1/\alpha}^{1/\beta} \frac{1}{z^2} f\left(\frac{1}{z}\right) dz + \int_{\alpha}^{\beta} f(x) dx
 \end{aligned}$$



Variable Transformation Example 1

$$\begin{aligned} I &= \int_0^{\infty} \frac{4}{\pi(4+x^2)} dx \\ &= \int_0^1 \frac{4}{\pi(4+x^2)} dx + \int_1^{\infty} \frac{4}{\pi(4+x^2)} dx \\ &= \int_0^1 \frac{4}{\pi(4+x^2)} dx - \int_1^0 \frac{1}{z^2} \frac{4}{\pi\left(4+\left(\frac{1}{z}\right)^2\right)} dz \\ &= \int_0^1 \frac{4}{\pi(4+x^2)} dx + \int_0^1 \frac{4}{\pi(4z^2+1)} dz \end{aligned}$$

Variable Transformation Example 2

$$\begin{aligned} I &= \int_{-\infty}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \int_{-\infty}^{-1} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx + \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= - \int_0^{-1} \frac{1}{z^2} \frac{1}{\sqrt{2\pi}} e^{-1/(2z^2)} dz + \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \int_{-1}^0 \frac{1}{\sqrt{2\pi} z^2} e^{-z^{-2}/2} dz + \int_{-1}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \end{aligned}$$

Open Formulas

- Open formulas are commonly used when the integrand blows up (singularity) at one or more end-points. e.g.,

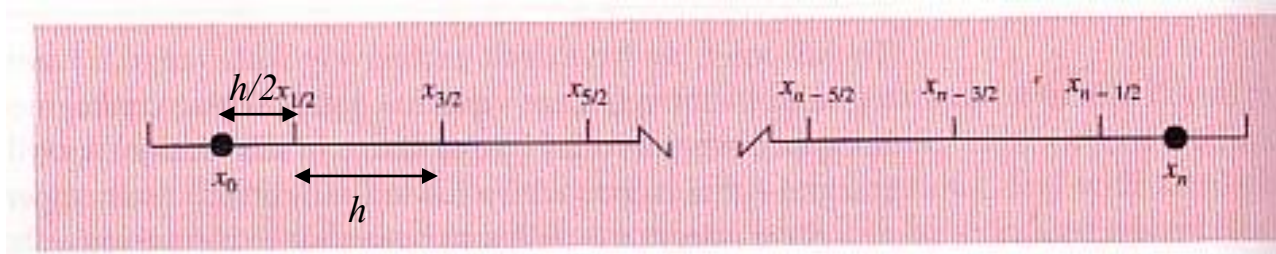
$$I = \int_0^1 \frac{1}{\sqrt{x}} dx$$

- Open formulas may be needed after variable transformation (but not always).
- The most common open formulas are mid-point rule and Gauss Quadrature.

Open Formulas

- Open formulas do not evaluate the integrand at the end points.
- Extended mid-point rule (see [mid_point_example.m](#)):

$$\int_{x_0}^{x_n} f(x)dx = h[f(x_{1/2}) + f(x_{3/2}) + \cdots + f(x_{n-3/2}) + f(x_{n-1/2})]$$



Improper Integrals: In-class exercises

- Apply mid-point rule to integrate (answer is 2):

$$I = \int_0^1 \frac{1}{\sqrt{x}} dx$$

- Split the following integral into two parts and then apply variable transformation and then evaluate using mid-point and/or Simpson rules (answers are 1 and 0.84):

$$I = \int_0^{\infty} \frac{4}{\pi(4+x^2)} dx$$

$$I = \int_{-\infty}^1 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

Improper Integral Examples

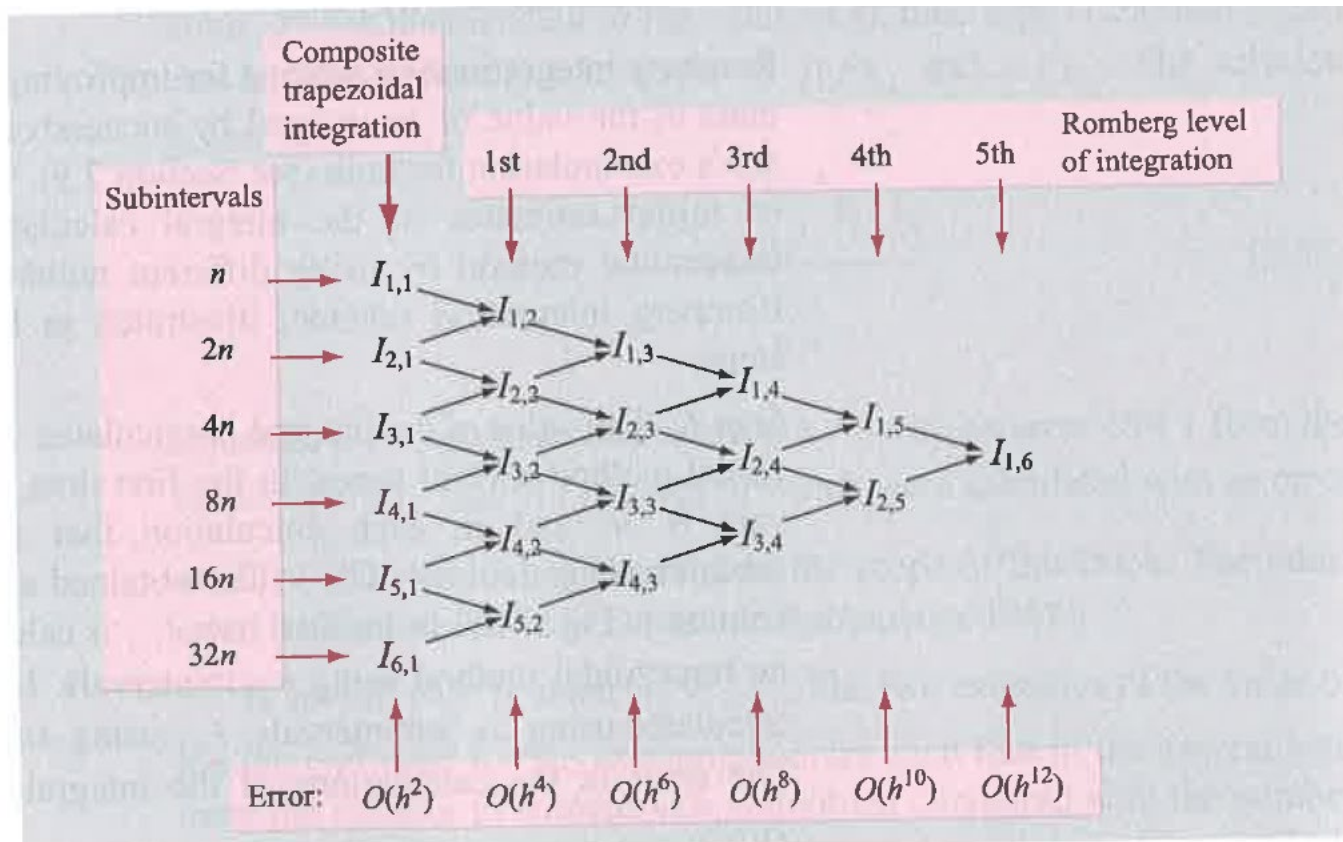
- The three examples included in the file below calculates open integrals using mid-point, gauss, and simpson rules.
- [open_integral_examples.m](#)

Romberg Integration

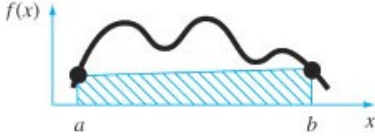
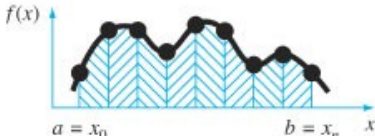
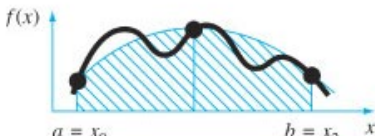
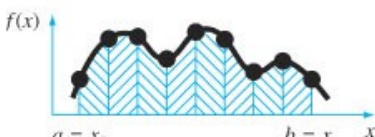
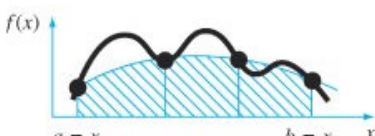
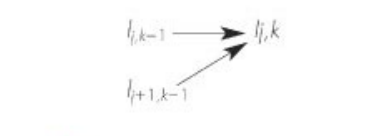
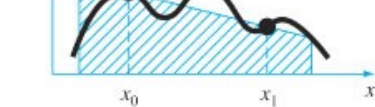
- Improved integral estimate can be obtained using trapezoidal estimates of the integral with multiple number of segments (n , $2n$, $4n$, $8n$, etc.), and then iteratively combining each pair using Richardson's extrapolation (add truncation error)
- Richardson's extrapolation formula
 - Two-level formula
$$I = \frac{4I(h) - I(2h)}{3}$$
 - Multi-level formula (i is level, j is iteration)

$$I = \frac{4^{j-1} I_{i+1,j-1} - I_{i,j-1}}{4^{j-1} - 1}$$

Romberg Integration



Numerical Integration Formulas

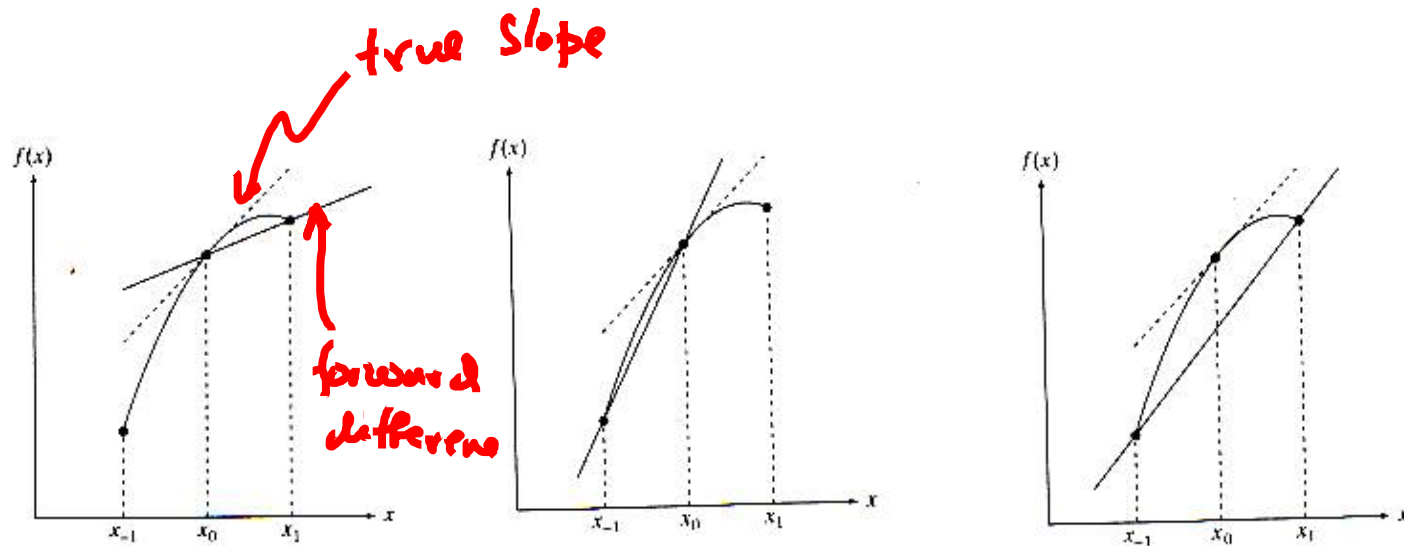
Method	Formulation	Graphic Interpretations	Error
Trapezoidal rule	$I \simeq (b-a) \frac{f(a) + f(b)}{2}$		$-\frac{(b-a)^3}{12} f''(\xi)$
Multiple-application trapezoidal rule	$I \simeq (b-a) \frac{f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}$		$-\frac{(b-a)^3}{12n^2} f''$
Simpson's 1/3 rule	$I \simeq (b-a) \frac{f(x_0) + 4f(x_1) + f(x_2)}{6}$		$-\frac{(b-a)^5}{2880} f^{(4)}(\xi)$
Multiple-application Simpson's 1/3 rule	$I \simeq (b-a) \frac{f(x_0) + 4 \sum_{i=1,3}^{n-1} f(x_i) + 2 \sum_{j=2,4}^{n-2} f(x_j) + f(x_n)}{3n}$		$-\frac{(b-a)^5}{180n^4} f^{(4)}$
Simpson's 3/8 rule	$I \simeq (b-a) \frac{f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)}{8}$		$-\frac{(b-a)^5}{6480} f^{(4)}(\xi)$
Romberg integration	$I_{i,k} = \frac{4^{k-1} I_{i+1,k-1} - I_{i,k-1}}{4^{k-1} - 1}$		$O(h^{2k})$
Gauss quadrature	$I \simeq c_0 f(x_0) + c_1 f(x_1) + \dots + c_{n-1} f(x_{n-1})$		$\simeq f^{(2n+2)}(\xi)$

NUMERICAL DIFFERENTIATION

Numerical Differentiation Outline

- Single Variable – Finite Differences
 - First Derivative, Second Derivative
 - First order, Higher order
- Multivariate
 - Gradient operator
 - Jacobian

Two-Point Finite-Differences



forward

backward

central

Finite Differences

- Difference formulas for numerical differentiation can be derived from Taylor series expansion.

$$f(x+h) = f(x) + f'(x)h + \frac{f''(x)h^2}{2} + \frac{f'''(x)h^3}{3!} \dots$$
$$f(x-h) = f(x) - f'(x)h + \frac{f''(x)h^2}{2} - \frac{f'''(x)h^3}{3!} \dots$$

- Using $x_{i+1} = x_i + h$ and $x_{i-1} = x_i - h$ we get

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2} + \frac{f'''(x_i)h^3}{6} \dots \quad (1)$$

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2} - \frac{f'''(x_i)h^3}{6} \dots \quad (2)$$

1st order formulas for 1st derivative

- Solving the first equation for $f'(x)$ and neglecting 2nd order terms we get the first order *forward* difference formula:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)$$

- Solving the 2nd equation for $f'(x)$ and neglecting 2nd order terms we get the first order *backward* difference formula:

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)$$

- Eliminating $f''(x)$ from equations (1) and (2) and then solving for $f'(x)$ we get the first order *central* difference formula:

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} + O(h^2)$$

Higher order formulas

- Accuracy of the numerical estimates of the derivative can be increased by using additional function values of the function farther away from the point of interest.
- Taylor series approximations for function values “ $2h$ ” distance away from x can be expressed as:

$$f(x+2h) = f(x) + f'(x)(2h) + \frac{f''(x)(2h)^2}{2} + \frac{f'''(x)(2h)^3}{3!} \dots$$

$$f(x-2h) = f(x) - f'(x)(2h) + \frac{f''(x)(2h)^2}{2} - \frac{f'''(x)(2h)^3}{3!} \dots$$

Higher order formulas

- The two point expansions can be summarized as:

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2} + \frac{f'''(x_i)h^3}{6} \dots \quad (1)$$

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2} - \frac{f'''(x_i)h^3}{6} \dots \quad (2)$$

$$f(x_{i+2}) = f(x_i) + 2f'(x_i)h + 2f''(x_i)h^2 + \frac{4f'''(x_i)h^3}{3} \dots \quad (3)$$

$$f(x_{i-2}) = f(x_i) - 2f'(x_i)h + 2f''(x_i)h^2 - \frac{4f'''(x_i)h^3}{3} \dots \quad (4)$$

2nd order formulas for the 1st derivative

- Eliminating $f''(x)$ from equations (1) and (3) and solving for $f'(x)$ we get the *forward* difference formula:

$$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h} + O(h^2)$$

- Eliminating $f''(x)$ from equations (2) and (4) and solving for $f'(x)$ we get the *backward* difference formula:

$$f'(x_i) = \frac{f(x_{i-2}) - 4f(x_{i-1}) + 3f(x_i)}{2h} + O(h^2)$$

2nd order formulas for the 1st derivative (contd.)

- Eliminating $f(x)$ and $f''(x)$ from equations (1),(2),(3) and (4) and solving for $f'(x)$ we get the *central* difference formula:

$$f'(x_i) = \frac{f(x_{i-2}) - 8f(x_{i-1}) + 8f(x_{i+1}) - f(x_{i+2}))}{12h} + O(h^4)$$

- In a slightly different compact notation ($i = 0$):

$$f'_0 = \frac{f_{-2} - 8f_{-1} + 8f_1 - f_2}{12h} + O(h^4)$$

General formulas for the 1st derivative

Weighting Coefficients and Truncation Error: First Derivative

	Points		Weighting Coefficients						Truncation	
Type	n		a_{-3}	a_{-2}	a_{-1}	a_0	a_1	a_2	a_3	Error
forward	2		0	0	0	-1	1	0	0	$-f^{(2)}(\xi)h/2$
	3		0	0	0	-3/2	2	-1/2	0	$f^{(3)}(\xi)h^2/3$
	4		0	0	0	-11/6	3	-3/2	1/3	$-f^{(4)}(\xi)h^3/4$
backward	2		0	0	-1	1	0	0	0	$f^{(2)}(\xi)h/2$
	3		0	1/2	-2	3/2	0	0	0	$f^{(3)}(\xi)h^2/3$
	4		-1/3	3/2	-3	11/6	0	0	0	$f^{(4)}(\xi)h^3/4$
central	3		0	0	-1/2	0	1/2	0	0	$-f^{(3)}(\xi)h^2/6$
	5		0	1/12	-2/3	0	2/3	-1/12	0	$f^{(5)}(\xi)h^4/30$

$$f'_0 = \frac{1}{h} \sum_{k=1-n}^{n-1} a_k f_k + O(h^{n-1})$$

1st order formulas for 2nd derivative

- Eliminating $f'(x)$ from equations (1) and (3) and solving for $f''(x)$ we get the first order forward difference formula for the 2nd derivative:

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + O(h)$$

- Eliminating $f'(x)$ from equations (2) and (4) and solving for $f''(x)$ we get the first order backward difference formula for the 2nd derivative:

$$f''(x_i) = \frac{f(x_{i-2}) - 2f(x_{i-1}) + f(x_i)}{h^2} + O(h)$$

1st order formulas for 2nd derivative (contd.)

- Eliminating $f'(x)$ from equations (1) and (2) and solving for $f''(x)$ we get the first order central difference formula for the 2nd derivative:

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2} + O(h^2)$$

- In a more compact notation ($i = 0$):

$$f_0'' = \frac{f_{-1} - 2f_0 + f_2}{h^2} + O(h^2)$$

General formulas for the 2nd derivative

Weighting Coefficients and Truncation Error: Second Derivative

Type	Points <i>n</i>	Weighting Coefficients							Truncation Error
		<i>b</i> ₋₃	<i>b</i> ₋₂	<i>b</i> ₋₁	<i>b</i> ₀	<i>b</i> ₁	<i>b</i> ₂	<i>b</i> ₃	
forward	3	0	0	0	1	-2	1	0	$-f^{(3)}(\xi)h$
	4	0	0	0	2	-5	4	-1	$11f^{(4)}(\xi)h^2/12$
backward	3	0	1	-2	1	0	0	0	$f^{(3)}(\xi)h$
	4	-1	4	-5	2	0	0	0	$11f^{(4)}(\xi)h^2/12$
central	3	0	0	1	-2	1	0	0	$-f^{(4)}(\xi)h^2/12$
	5	0	-1/12	4/3	-5/2	4/3	-1/12	0	$f^{(6)}(\xi)h^4/90$

$$f_0'' = \frac{1}{h^2} \sum_{k=1-n}^{n-1} b_k f_k + O(h^{n-2})$$

Summary of Forward Difference Formulas

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$$

Error

$$O(h)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$$

$$O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2}$$

$$O(h)$$

$$f''(x_i) = \frac{-f(x_{i+3}) + 4f(x_{i+2}) - 5f(x_{i+1}) + 2f(x_i)}{h^2}$$

$$O(h^2)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+3}) - 3f(x_{i+2}) + 3f(x_{i+1}) - f(x_i)}{h^3}$$

$$O(h)$$

$$f'''(x_i) = \frac{-3f(x_{i+4}) + 14f(x_{i+3}) - 24f(x_{i+2}) + 18f(x_{i+1}) - 5f(x_i)}{2h^3}$$

$$O(h^2)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+4}) - 4f(x_{i+3}) + 6f(x_{i+2}) - 4f(x_{i+1}) + f(x_i)}{h^4}$$

$$O(h)$$

$$f^{(4)}(x_i) = \frac{-2f(x_{i+5}) + 11f(x_{i+4}) - 24f(x_{i+3}) + 26f(x_{i+2}) - 14f(x_{i+1}) + 3f(x_i)}{h^4}$$

$$O(h^2)$$

Summary of Backward Difference Formulas

First Derivative

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$$

Error

$$O(h)$$

$$f'(x_i) = \frac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{2h}$$

$$O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2}$$

$$O(h)$$

$$f''(x_i) = \frac{2f(x_i) - 5f(x_{i-1}) + 4f(x_{i-2}) - f(x_{i-3}))}{h^2}$$

$$O(h^2)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_i) - 3f(x_{i-1}) + 3f(x_{i-2}) - f(x_{i-3}))}{h^3}$$

$$O(h)$$

$$f'''(x_i) = \frac{5f(x_i) - 18f(x_{i-1}) + 24f(x_{i-2}) - 14f(x_{i-3}) + 3f(x_{i-4}))}{2h^3}$$

$$O(h^2)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_i) - 4f(x_{i-1}) + 6f(x_{i-2}) - 4f(x_{i-3}) + f(x_{i-4}))}{h^4}$$

$$O(h)$$

$$f^{(4)}(x_i) = \frac{3f(x_i) - 14f(x_{i-1}) + 26f(x_{i-2}) - 24f(x_{i-3}) + 11f(x_{i-4}) - 2f(x_{i-5}))}{h^4}$$

$$O(h^2)$$

Summary of Central Difference Formulas

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

Error

$$O(h^2)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h}$$

$$O(h^4)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

$$O(h^2)$$

$$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2}))}{12h^2}$$

$$O(h^4)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2}))}{2h^3}$$

$$O(h^2)$$

$$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3}))}{8h^3}$$

$$O(h^4)$$

Fourth Derivative

$$f^{(4)}(x_i) = \frac{f(x_{i+2}) - 4f(x_{i+1}) + 6f(x_i) - 4f(x_{i-1}) + f(x_{i-2}))}{h^4}$$

$$O(h^2)$$

$$f^{(4)}(x_i) = \frac{-f(x_{i+3}) + 12f(x_{i+2}) + 39f(x_{i+1}) + 56f(x_i) - 39f(x_{i-1}) + 12f(x_{i-2}) + f(x_{i-3}))}{6h^4}$$

$$O(h^4)$$

Differentiation of Tabular Data

- Differentiation of tabular data (unequally spaced) can be handled by direct forward or central differencing if the point of interest lies within the data limits
- E.g., forward differencing

$$f'(x) = \frac{f(x_{i+1}) - f(x_i)}{(x_{i+1} - x_i)}, \quad x_i < x < x_{i+1}$$

- If the derivatives are desired at all given points, forward differencing may be used for the first point, central differencing for the interior points, and backward differencing for the last point (`gradient` function in Matlab)

Differentiation of Tabular Data

- Differentiation of tabular data (unequally spaced) can be handled by fitting an interpolating polynomial (Newton or Lagrange – to be covered later) to the data and then analytically differentiating it.
- The following expression evaluates the derivative at point 'x' using values at 3 adjacent points (from 2nd order Lagrange interpolating polynomial):

$$f'(x) = f(x_{i-1}) \frac{2x - x_i - x_{i+1}}{(x_{i-1} - x_i)(x_{i-1} - x_{i+1})} + f(x_i) \frac{2x - x_{i-1} - x_{i+1}}{(x_i - x_{i-1})(x_i - x_{i+1})} \\ + f(x_{i+1}) \frac{2x - x_{i-1} - x_i}{(x_{i+1} - x_{i-1})(x_{i+1} - x_i)}$$

Accuracy of Finite Differences

- Numerical accuracy of finite-differences depend on many factors.
- Central difference is generally more accurate than either forward or backward difference.
- Increase in number of points n leads to higher accuracy (lower truncation error).
- Smaller values of h leads to lower truncation error but too small a value may result in numerical roundoff errors.
- Richardson's extrapolation may be used to improve accuracy of finite-differences.

Differentiation Examples

- Write MATLAB code to perform central differencing to compute first and second derivatives.
- Rewrite so that the first derivative function can be recursively called to evaluate the n^{th} derivative.
- Examples:
 - [simple differentiation examples.m](#)

Jet Fighter example

(C 21.12, p. 541 or C&C 24.44, p. 691)

- Jet fighter landing on aircraft carrier

t (s)	0	0.52	1.04	1.75	2.37	3.25	3.83
x (m)	153	185	210	249	261	271	273

- Develop code for differentiation of tabular data and find velocity $v = dx/dt$ and acceleration $a = dv/dt$.

Aircraft Radar Tracking example (C & C 23.22 p. 670)

- Position of plane tracked by radar

T (s)	200	202	204	206	208	210
Θ (rad)	0.75	0.72	0.70	0.68	0.67	0.66
r (m)	5120	5370	5560	5800	6030	6240

- Find velocity and acceleration vectors and plot against time

$$\vec{v} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{e}_\theta$$

- What are the values at t = 206

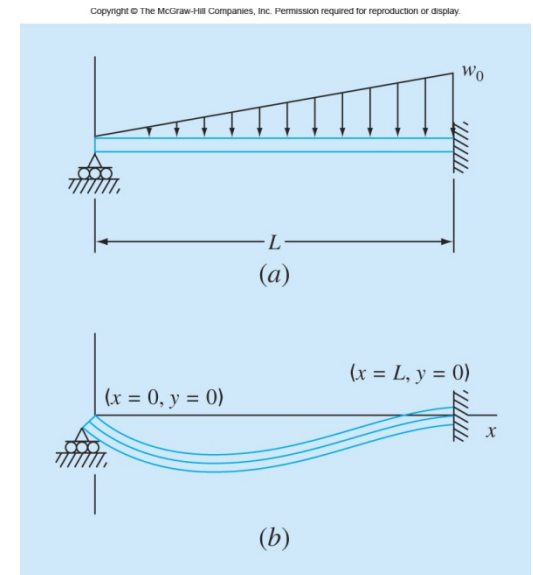
Beam example

(C & C 24.19-24.20, p. 685)

- Given deflection data find slope θ , moment M , shear V , and distributed load w :

$$\frac{dy}{dx} = \theta(x); \quad \frac{d\theta}{dx} = \frac{M(x)}{EI}; \quad \frac{dM}{dx} = V(x); \quad \frac{dV}{dx} = -w(x)$$

- $E=200$ Gpa, $I = 0.0003$ m⁴, $w_0 = 2.5$ kN/cm, $dx = 0.125$ m, $L = 3$ m



X (m)	0	0.375	0.75	1.125	1.5	1.875	2.25	2.625	3
Y (cm)	0	-0.2571	-0.9484	-1.9689	-3.2262	-4.6414	-6.1503	-7.7051	-9.275

Richardson's Extrapolation

- Accuracy of finite-differencing can be improved by error-correction (Richardson's extrapolation) if two estimates are available at different h values.
- If we choose these different h values to be h and $2h$, then the k^{th} derivative can be expressed as a linear combination of the two n point estimates (Shilling and Harris)

$$\bar{f}_0^{(k)} = \frac{2^{n-k} f_0^{(k)}(h) - f_0^{(k)}(2h)}{2^{n-k} - 1}$$

- For the special case, $n = 3$ and $k = 1$ (first order central differencing)

$$D = \frac{4D(h) - D(2h)}{3}$$

Multivariate Differentiation

- Multivariate differentiation is performed using partial derivatives.
- The vector operator for multivariate differentiation is the gradient or del (∇) operator and it generally yields vectors if we have a single function of multiple variables (e.g., $f(x, y)$).

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \nabla f \cdot d\mathbf{r}$$

$$\begin{aligned} \frac{\partial f(x, y)}{\partial x} &= \frac{d}{dx} (f(x, y)) \Big|_{y=\text{const}} \\ &= \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \end{aligned}$$

Computation of Gradient Vector

- If the gradient vector $\nabla f(\mathbf{x})$ is computed using central finite differences, $2n$ function evaluations may be required.

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{f(x_1 + \delta x_1, \dots, x_n) - f(x_1 - \delta x_1, \dots, x_n)}{2\delta x_1} \\ \vdots \\ \frac{f(x_1, \dots, x_j + \delta x_j, \dots, x_n) - f(x_1, \dots, x_j - \delta x_j, \dots, x_n)}{2\delta x_j} \\ \vdots \\ \frac{f(x_1, \dots, x_n + \delta x_n) - f(x_1, \dots, x_n - \delta x_n)}{2\delta x_n} \end{pmatrix}$$

Jacobian Matrix

- If we have a vector of functions (multiple functions) then the gradient operator yields a matrix called the Jacobian matrix.
- If possible, the Jacobian matrix should be computed analytically.
- Finite differencing can be used if the function $\mathbf{F}$ is not an analytical function (e.g., discrete approximation to a complicated PDE).

Jacobian Matrix Computation

$$\mathbf{J}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_j} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \dots & \vdots & \dots & \vdots \\ \frac{\partial f_i}{\partial x_1} & \dots & \frac{\partial f_i}{\partial x_j} & \dots & \frac{\partial f_i}{\partial x_n} \\ \vdots & \dots & \vdots & \dots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_j} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

Jacobian Matrix Computation using central finite differences

$$\frac{\partial f_i}{\partial x_j} = \frac{\partial f_i(x_1, x_2, \dots, x_j, \dots, x_n)}{\partial x_j} =$$
$$\frac{f_i(x_1, x_2, \dots, x_j + \delta x, \dots, x_n) - f_i(x_1, x_2, \dots, x_j - \delta x, \dots, x_n)}{2\delta x}$$

Differentiation using MATLAB functions

- Matlab function `diff` can be used for differentiation.
- If it is an analytical function then `diff` should be used in symbolic mode.
- Matlab function `gradient` can be used to compute the gradient.
- Symbolic matlab function `jacobian` can be used to compute the Jacobian matrix.

Multivariate Examples

- Write a function to compute the numerical gradient of a multivariate function at any given point.
- Write a function to compute the Jacobian of a set of multivariate functions at any given point.
- Example:
 - [multivariate differentiation examples.m](#)

Handwritten SLiDEs

Types of Integrands

- Simple continuous function
- Complicated continuous function
- Tabulated function or data

A hand-drawn diagram illustrating the components of an integral. On the left, a large circle contains a red letter 'I', with the word 'Integral' written below it. This is followed by an equals sign. To the right of the equals sign is a red integral symbol. A red arrow points from the word 'Integrand function' to the red 'f(x)' inside the integral. Another red arrow points from the word 'limits' to the red 'a' at the bottom of the integral. A third red arrow points from the word 'independent variable' to the red 'x' inside the differential 'dx'. The entire expression is written in red ink.

$$I = \int_a^b f(x) dx$$

Integral

Integrand function

limits

independent variable

$$d = \int_0^T v(t) dt$$

$$\frac{dv}{dt} = g - \frac{c}{m} v^2$$

$$v(t) = \sqrt{\frac{gm}{c}} \tanh\left(\sqrt{\frac{gc}{m}} t\right)$$

Bungee Jumper Line Integral

$$d = \int \underbrace{\sqrt{\frac{gm}{c}} \tanh\left(\sqrt{\frac{gc}{m}} t\right)}_{v(t)} dt$$
$$\Rightarrow \frac{m}{c} \ln \left[\cosh \left(\sqrt{\frac{gc}{m}} t \right) \right]$$

Tabular Data Example 1

Chapra 19.8 (p. 493)

Calculate 'd'

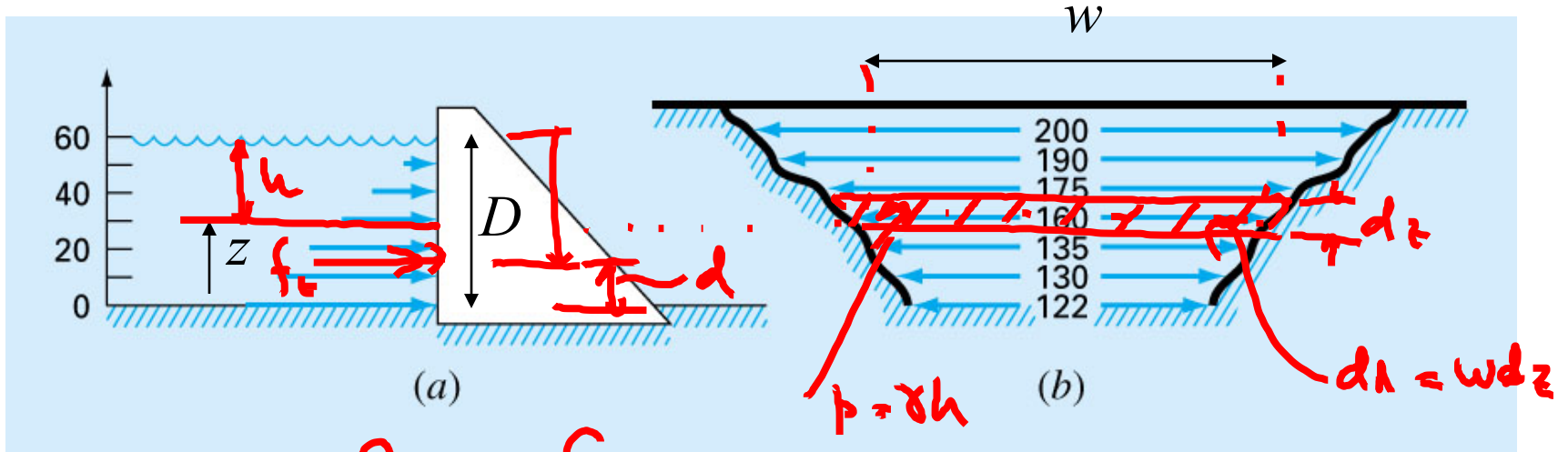
$$d = \int_0^{10} v(t) dt$$

t	1	2	3.25	4.5	6	7	8	8.5	9	10
v	5	6	5.5	7	8.5	8	6	7	7	5

① Trapezoidal rule

② Fit data with a cubic polynomial
and integrate

Tabular Data Example 3 (C&C 24.23, p 686)



$$F = \int p \, dA = \int_{\text{bottom}} \rho g h w \, dz$$

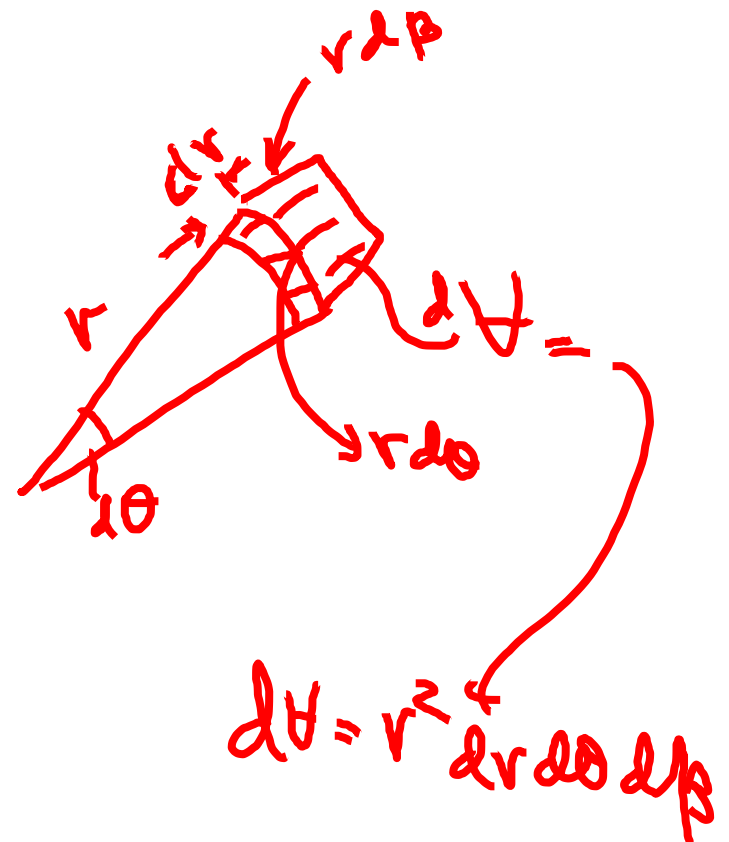
$$f_t = \int_0^D \rho g(D-z)w(z)dz$$

$$d = \frac{\int_0^D \rho g z (D - z) w(z) dz}{\int_0^D \rho g (D - z) w(z) dz}$$

Mass of Earth Example

$$M = \int_0^R \int_0^{2\pi} \int_0^{2\pi} \rho dV$$

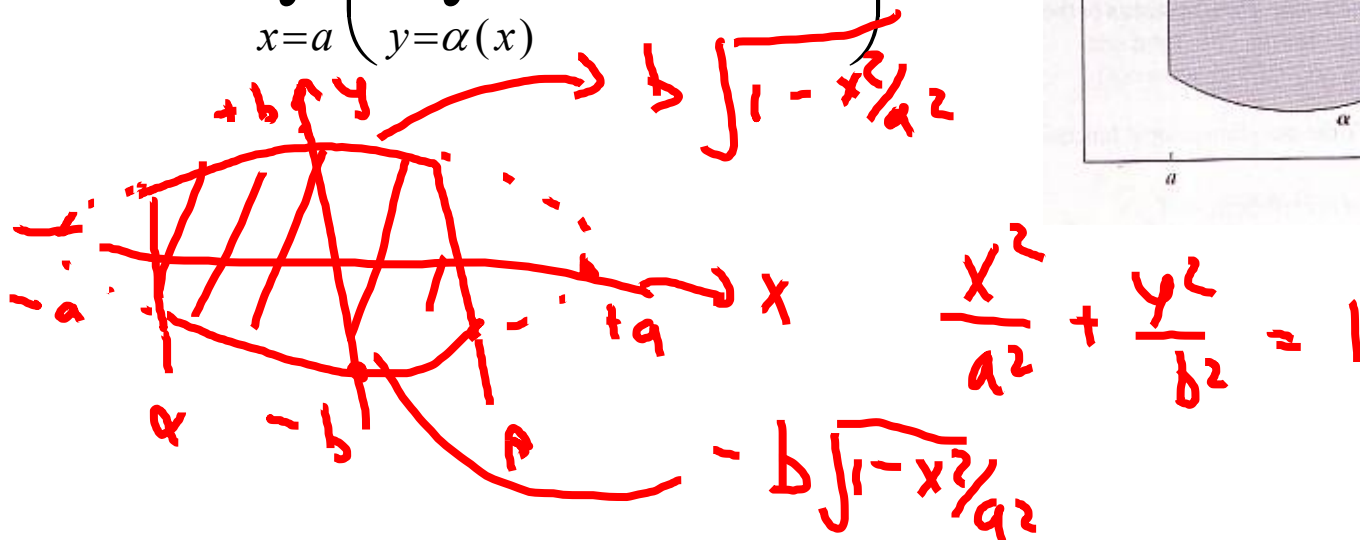
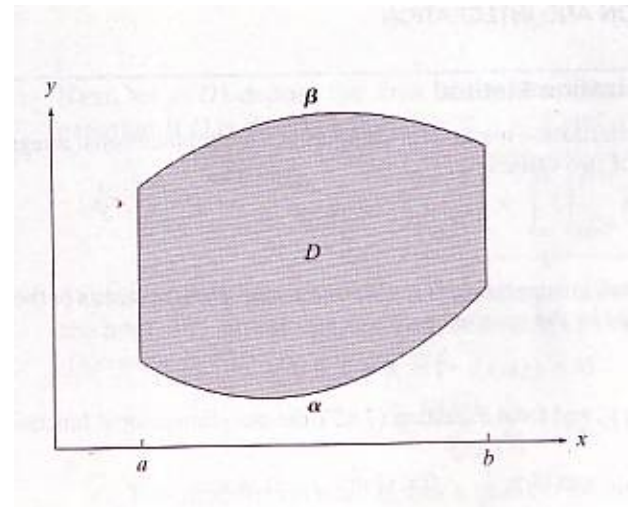
r	0	...	6380
ρ	13	...	3



Extension to Variable Limits

- Multiple integrals can sometimes have inner limits expressed as functions of the outer variable:

$$I = \int_{x=a}^{x=b} \left(\int_{y=\alpha(x)}^{y=\beta(x)} f(x, y) dy \right) dx$$



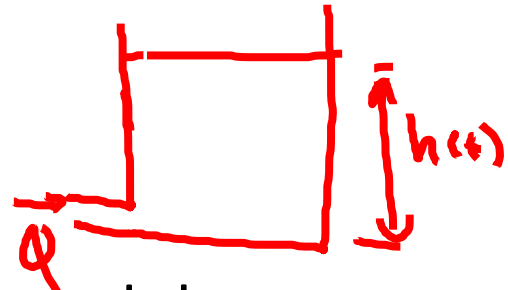
$$\int_a^b \int_{-b\sqrt{1-x^2/a^2}}^{b\sqrt{1-x^2/b^2}} dy \, dx$$

In class Example

- Extend previous tank example with time dependent change in fluid volume
- Assume there is a volumetric influx of liquid into the tank at the following sinusoidal rate:

$$V = \int_0^t Q dt$$

$$Q = Q_0 (1 + \sin(\omega t))$$



- Assume the tank is empty in the beginning and the concentration distribution is the same as before
- Use $Q_0 = 100 \text{ L/s}$, $\omega = \pi/6$, t is in seconds
- Calculate total mass at $t = 1 \text{ hour}$

$$h(t) = \frac{V(t)}{A}$$

$$M = \int_0^H \int_{-b}^b \int_{-a}^a \frac{\rho_0 (x^2 + y^2)}{\sqrt{1+z^2}} dx dy dz$$

$$H = \int_0^T \varphi(t) dt / \lambda$$

In class Example (contd.)

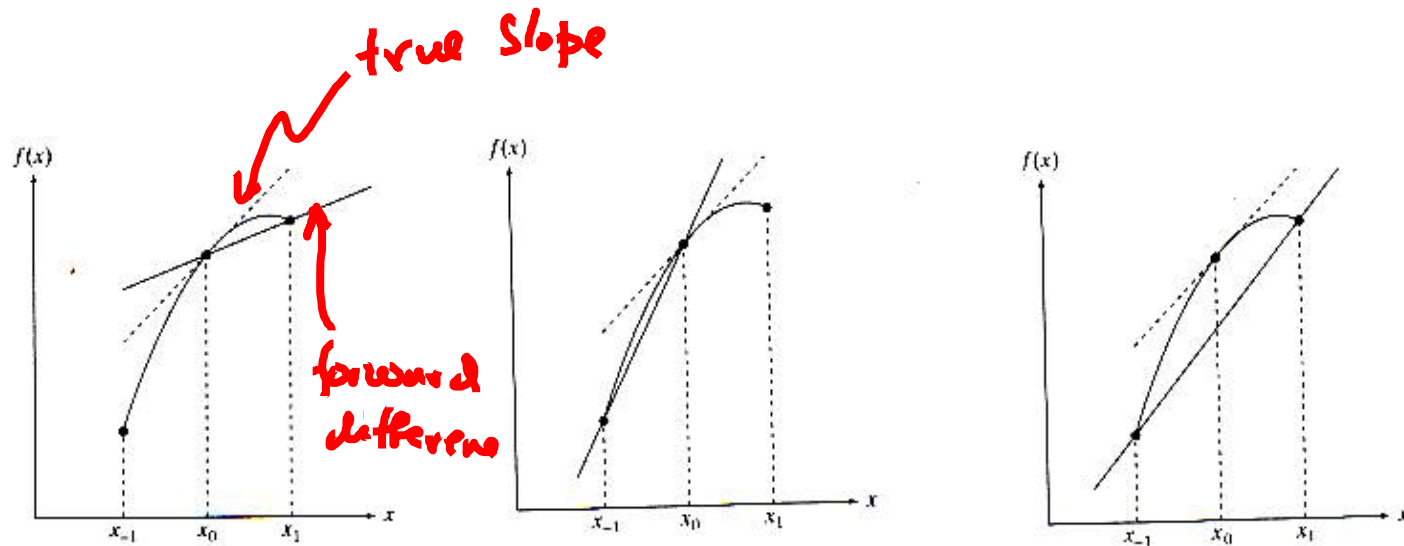
- Extend previous example for time dependent concentration in tank with the following distribution

$$c(x, y, z, t) = \frac{C_0 e^{-kt} (x^2 + y^2)}{\sqrt{z^2 + 1}}$$

$$M = \int_0^T \int_0^b \int_{-b}^b \int_{-a}^a$$

- Assume the tank is empty in the beginning
- Use $k = 0.001$, all other parameters remain same
- Plot total mass (M) vs time (t) for $t = 0$ to 3600 s
- Calculate the time averaged mass in the tank over this period

Two-Point Finite-Differences



forward

backward

central

1st order formulas for 2nd derivative (contd.)

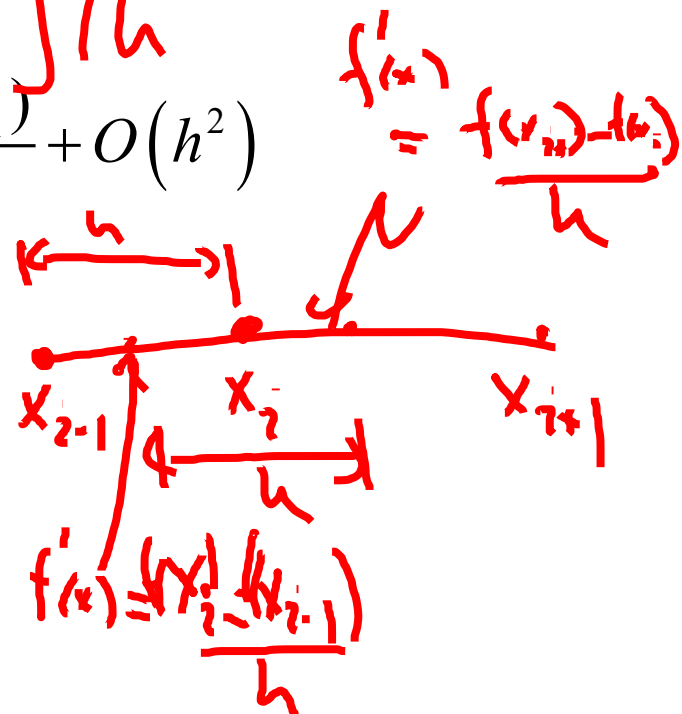
- Eliminating $f'(x)$ from equations (1) and (2) and solving for $f''(x)$ we get the first order central difference formula for the 2nd derivative:

$$f''(x) = \left[\frac{f(x_{i+1}) - f(x_i)}{h} - \frac{f(x_i) - f(x_{i-1}))}{h} \right] / h$$

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2} + O(h^2)$$

- In a more compact notation ($i = 0$):

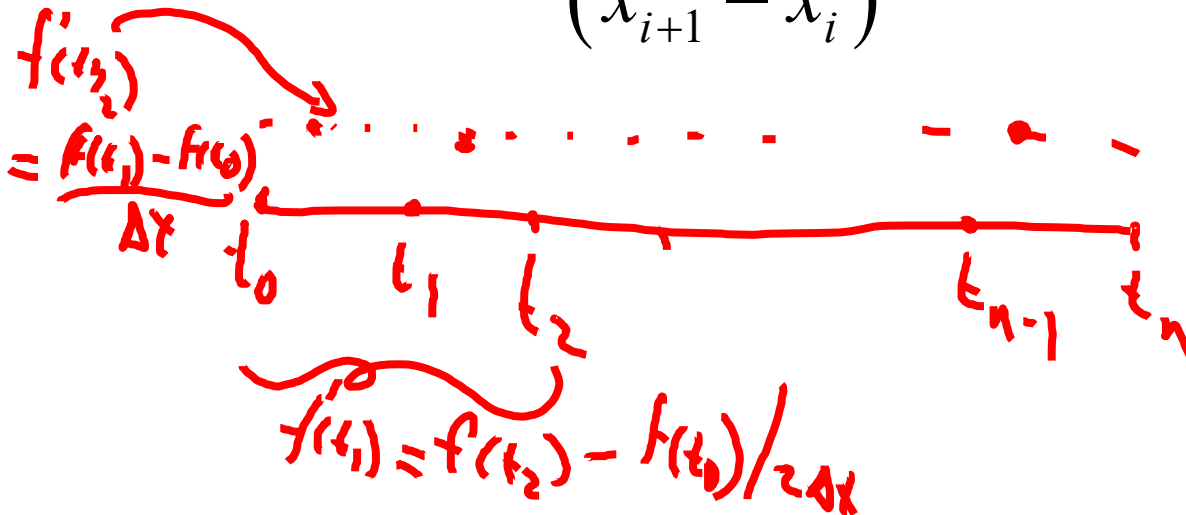
$$f''_0 = \frac{f_{-1} - 2f_0 + f_2}{h^2} + O(h^2)$$



Differentiation of Tabular Data

- Differentiation of tabular data (unequally spaced) can be handled by direct forward or central differencing if the point of interest lies within the data limits:

$$f'(x) = \frac{f(x_{i+1}) - f(x_i)}{(x_{i+1} - x_i)}, \quad x_i < x < x_{i+1}$$



Chapra 21.12 (p.540)

t	0	0.52	1.04	...	3.83
x	153	185	208		273

$$V = \frac{dx}{dt}, \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

Multivariate Differentiation

- Multivariate differentiation is performed using partial derivatives.
- The vector operator for multivariate differentiation is the gradient or del (∇) operator and it generally yields vectors if we have a single function of multiple variables (e.g., $f(x, y)$).

$$\nabla f = \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \end{pmatrix}$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \nabla f \cdot d\mathbf{r}$$

$$\begin{aligned} \frac{\partial f(x, y)}{\partial x} &= \frac{d}{dx} (f(x, y)) \Big|_{y=\text{const}} \\ &= \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \end{aligned}$$

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} ()$$